

Machine Learning Classification:
Differentiating Bacterial Exotoxins based on
Exotoxin Types and Targets

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March 4, 2025

Acknowledgements

As of today, I am completing this master's thesis at LMU as a master's student at the Technical University of Munich (TUM); I want to thank everyone who participated in my walking this path.

I sincerely thank Dr. Luisa Jiménez-Soto for her unwavering support and guidance. Her vision and encouragement have been essential in shaping this thesis project. I will never forget our scientific discussions, which has expanded my research perspective day by day.

I also sincerely want to thank Prof. Dr. Aurélien Tellier for examining this thesis and giving me valuable feedback on my thesis defence. Thanks to his trust and constructive criticism, I was able to grow academically during my thesis period.

I also want to thank my colleagues at the Walther Straub Institute of Pharmacology and Toxicology. I am particularly thankful to Tanja Krüger; her mentorship and humour have made even the most challenging days more manageable. I also want to thank Sudenaz Yakın, Lea Bernhart, Franzi Silgoner, and Sebastian Franz for making this experience enjoyable.

Furthermore, I am grateful to my family for their support. To my beloved parents, my sisters Ayça and Destan, my grandmother—whose academic insights have always been explicit—and my dear Asu, I want to thank you all for your support and love.

I also want to sincerely thank my friends, Samed Bayer, Betülai Kuş, and Ayşe Akalın, as they have been a constant source of motivation and comfort throughout this journey.

Abstract

The rapid expansion of sequencing data in genomics and healthcare presents both opportunities and challenges, particularly in the study of bacterial exotoxins. Since the early 2000s, high-throughput sequencing has generated vast amounts of unlabeled genomic data, including sequences encoding exotoxins. Bacterial exotoxins play key roles in therapeutics, such as pain management, bacterial infection treatments, and neurodegenerative disease research. However, experimental (wet lab) research on exotoxins is labor-intensive and time-consuming, creating a ~~need~~ opportunity for computational approaches ~~for~~ to further research faster.

Machine learning has emerged as a powerful tool for analyzing these complex sequences, but existing classification models primarily focus on binary distinctions (e.g., toxin vs. non-toxin, exotoxin vs. endotoxin). This study addresses a critical gap by developing predictive models that classify exotoxins by functional type and target specificity (vertebrate vs. invertebrate).

Using a non-redundant dataset of 1,067 sequences, we developed ExoType and ExoTarget, two machine learning models leveraging protein embeddings generated by large language models with T5 architecture. ExoType, an SVM-based predictor, classifies exotoxins based on their functional type, while ExoTarget, a Random Forest model, predicts their biological targets. Both models integrate sequence-derived embeddings (MMseqs2) and structure-based embeddings (AlphaFold2) to improve classification performance.

Our models achieved Matthews Correlation Coefficients (MCC) of 70% for ExoType and 67% for ExoTarget, demonstrating the potential of machine learning and large language models in advancing bacterial exotoxin classification.

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Chapter 1

Introduction

1.1 Bacterial Exotoxins

Bacterial toxins are molecules produced by bacteria that can cause harm by interacting with host tissues or disrupting cellular processes. Some bacteria secrete extracellular enzymes that degrade host tissue or alter immune responses. [1]. For example, diphtheria toxin prevents protein synthesis. Similarly, *S. aureus* can overstimulate immune responses, while cholera toxin modifies second messenger pathways to aid bacterial survival and transmission without directly invading host tissues. [2].

Bacterial toxin forms are classified into bacterial endotoxins and exotoxins. Endotoxins are known as lipopolysaccharides (LPS) found in the outer membrane of Gram-negative bacteria and are released upon bacterial cell lysis. Exotoxins, on the other hand, are proteins actively secreted into the extracellular environment by both Gram-positive and Gram-negative bacteria [3]. Therefore, exotoxins have several modes of action that allow the bacteria to manipulate the host.

Bacterial exotoxins are primarily classified as Type I, Type II, and Type III; although a fourth type has been proposed, it has not yet been fully recognized in the scientific literature. Type I exotoxins, mainly superantigens, do not enter host cells. Instead, they overstimulate the immune system by triggering excessive cytokine release, leading to toxic shock syndrome. Type II exotoxins, known as membrane-disrupting toxins, function in two main ways. They either form pores in host cell membranes, causing osmotic imbalance and lysis, or they enzymatically degrade membrane phospholipids,

particularly in immune cells such as phagocytes. Type III toxins, commonly referred to as A-B toxins, composed of an A subunit that disrupts intracellular processes and a B subunit that facilitates toxin entry by binding to specific host receptors. The proposed Type IV exotoxins operate by degrading extracellular matrix. In general, exotoxins contribute significantly to bacterial virulence by impairing host immune responses, damaging tissues, and facilitating bacterial survival and proliferation [4] [5]. Their biosynthesis can be influenced by a variety of environmental and genetic factors[6].

1.2 Motivation

The emergence of high-throughput technologies has improved bacterial toxin research by reducing the cost of sequencing bacterial genomes. As bacterial genomes are being sequenced at an accelerating rate, the need for efficient annotation tools has emerged.[7][8] While trying to annotate a vast amount of sequences, wet-lab methods have been inadequate due to being time-consuming, costly and error-prone. Therefore, machine learning offers a promising computational alternative to overcome these limitations. Machine learning uses algorithms to get trained from data by recognizing distinct patterns in it [9]. Data can be either unlabeled, which is used for unsupervised learning, or labeled, which makes it suitable for supervised learning.[10]. A method used for supervised learning is classification which divides the data in two classes. Traditional classification methods, a type of supervised learning, can be designed for binary or multiclass problems. The classification of bacterial exotoxins has drawn attention in recent years, mainly with the use of machine learning (ML) techniques.[11].

Several studies have leveraged machine learning for toxin classification, utilizing diverse models and methodologies [12][13]. However, most of them have focused on general bacterial classification. They have also highlighted challenges for exotoxin classification, including small dataset sizes, issues with image quality, and the need for advanced feature extraction techniques [14].

Over the years, there still remains a significant gap in understanding if accurately distinguishing bacterial exotoxin among all types is possible. Machine learning classification methods have primarily relied on binary classifications such as toxin versus non-toxin or exotoxin versus endotoxin. [12][13][14]. Although current exotoxin research has been essential for understanding bac-

terial virulence factors, these methods often fail to provide insights into the distinct types of exotoxins based on their four mechanisms of action.

Furthermore, the research community has not yet developed any predictor that distinguishes bacterial exotoxins based on the organism it targets and invades. Motivated by these gaps in the existing literature, this study aims to fill this gap by building novel predictors: ExoType and ExoTarget. ExoType predictors aim to distinguish among five types of exotoxins based on two types of inputs: protein sequences and protein folds. On the other hand, ExoTarget predictors are designed to classify the different targets of exotoxins based on two types of inputs: protein sequences and protein folds.”

1.3 Objectives

This study encompasses three objectives. The first objective is to develop ExoType, a predictor for classifying bacterial exotoxin types based on their mechanisms of action. It reviews five types of bacterial exotoxins that were selected based on their types: membrane-transducing (Type I), pore-forming (Type II), intracellular toxins (Type III), ECM-degrading (Type IV), and other types. The second is to predict the classification of bacterial exotoxin targets by building ExoTarget, if possible. This encompasses differentiating between targets of bacterial exotoxins, categorizing them into vertebrate and non-vertebrate targets. The final one is to inspect the best performing machine learning model for the classification of bacterial toxins based on types and targets. This objective investigates whether training models on bacterial exotoxin types and targets can achieve a competitive predictive performance.

1.4 Structure of Thesis

- Chapter 1 introduces the key terms, outlines the motivation behind the thesis project, and provides an overview of the thesis structure.
- Chapter 2 provides background information, which includes a comprehensive review of bacterial exotoxins, machine learning classification systems and foundational machine learning techniques relevant to this study. It also reviews contributions to computational applications of bacterial exotoxins and explains how this study addresses a current gap in the field.

- Chapter 3 details the methodologies employed in this study. It covers data collection, preprocessing, feature extraction, model development, hyperparameter tuning, and model evaluation.
- Chapter 4 presents the results gathered from the feature analysis of protein sequences, protein embeddings, machine learning models including comparative performance metrics and visualizations.
- Chapter 5 analyzes the findings, compares them with existing literature, and explores their implications for machine learning applications in bacterial exotoxin research.
- Chapter 6 details the summary of the thesis project, its limitations, and discusses possible future work.

Chapter 2

Background

Chapter 3

Methods

~~This study applies~~ In this study, I apply machine learning techniques to classify bacterial exotoxins based on their functional types and biological targets. The methodology follows a structured pipeline that involves data collection, pre-processing, feature extraction, and model training.

First, primary sequences ~~was~~ were collected and analyzed using descriptive statistics to understand key sequence characteristics. Next, the dataset was preprocessed using redundancy reduction techniques to generate a non-redundant set of sequences. Next, protein language models were used to generate numerical embeddings that capture meaningful sequence representations.

Then, dimensionality reduction using Principal Component Analysis is applied to the embeddings. The processed embeddings are then used to train machine learning models, including conventional classifiers and hierarchical predictors. The machine learning models are evaluated using performance metrics for binary and multiclass classification tasks.

Figure 3.1 visualizes the key steps in building a machine learning classification pipeline for differentiating exotoxin types and targets.

3.1 Data Collection and Curation

Bacterial toxins protein sequences were collected from public sequence databases, such as UniProt and NCBI, as described previously by Krueger et al [15]. The dataset was curated and labeled by selecting proteins from bacterial pathogens based on their host specificity by PD. Dr. Luisa F. Jimenez-Soto.

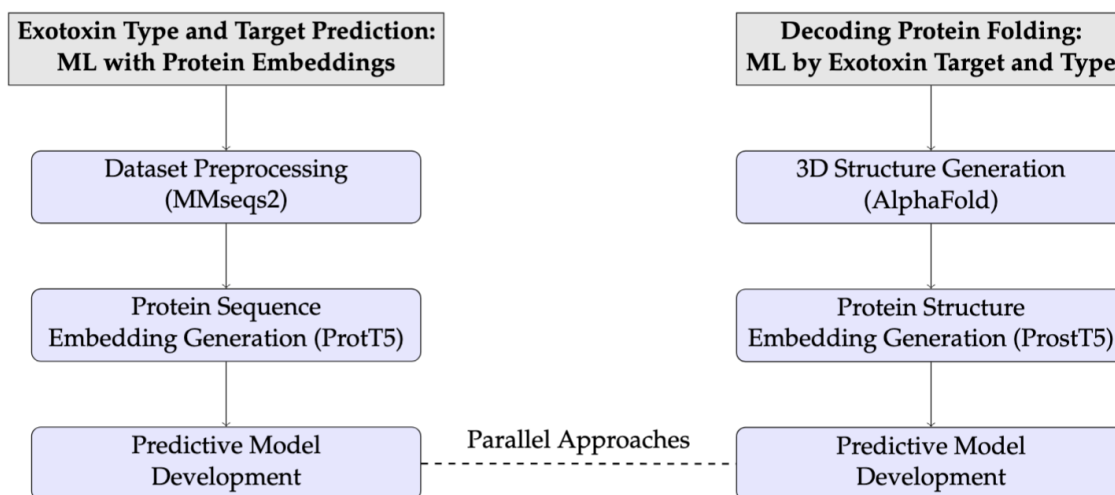


Figure 3.1: Overview of the key steps in the methodology: Exotoxin type and target predictions based on protein (left) and protein folding-based classification (right).

Proteins from bacterial pathogens affecting insects, plants, and other bacteria were categorized as non-vertebrate targets, whereas proteins from bacterial pathogens affecting vertebrates—including fish, amphibians, birds, reptiles, and mammals—were classified as vertebrate targets.

Similarly, bacterial exotoxins were categorized based on their mode of action on how they exert their effect on the host. Bacterial exotoxins were classified into different functional types, including membrane-transducing toxins (Type I), pore-forming toxins (Type II), intracellular toxins (Type III), and toxins that degrade the extracellular matrix (Type IV).

3.1.1 Dataset Structure and Pre-processing

The curated dataset was structured in a **CSV** *csv* file, which contains bacterial species, their exotoxin types, and their specific targets. This labeled data was used to annotate the input files to build a predictive model for classifying bacterial exotoxins.

usually for file types, only lowercase are used. That is why I changed the CSV to csv. Check the literature before, but I have seen them only in lowercase

The FASTA file stores each exotoxin ID alongside its corresponding protein sequence. To integrate these datasets, the CSV and FASTA files were merged using their IDs. The resulting fully labeled dataset includes bacterial exotoxin sequences along with their respective types and targets. This data preprocessing step is implemented in the `data-preprocessing.py` script using Python 3.13.1.

Then, a Python script was employed to eliminate sequences annotated as 'partial' or 'fragment' in their FASTA headers, thereby reducing incomplete sequences.

3.1.2 Dataset Composition

The final dataset contains 2,473 sequences. Type III toxins have 2,148 entries, including both Type III and Type III Control. Type II toxins have 169 entries, including Type II and Type II PFT bacteria. Type I toxins have 42 entries, while Type IV toxins have 28 entries. An additional 104 entries are categorized as Unknown, as they do not belong to any of the four existing classes.

The target category consist of vertebrate-targeting exotoxins and non-vertebrate-targeting bacterial exotoxins, with 1,863 and 629 ~~data-points~~ ~~sequences~~, respectively.

3.2 Descriptive Statistical Analysis of Sequences

Understanding the statistical properties of protein sequences is essential for analyzing their biological characteristics and identifying potential biases in the dataset. Descriptive statistical analysis provides insights into fundamental sequence properties, such as length distributions and hydrophobicity.

This analysis was conducted using custom Python scripts, incorporating key libraries such as Biopython ~~1.85-~~ (version 1.85) for sequence handling, NumPy ~~2.2.2~~ (version 2.2.2) for numerical computations, and Matplotlib ~~3.10.0~~ version 3.10.0 for visualization.

3.2.1 Mean Sequence Length

Understanding length variability is crucial, as it may influence the training of machine learning models. The mean sequence length μ was calculated by

using the following formula:

$$\mu = \frac{1}{N} \sum_{i=1}^N L_i \quad (3.1)$$

where:

- μ is the mean sequence length,
- N is the total number of sequences in the dataset,
- L_i represents the length of the i^{th} sequence.

3.2.2 Hydrophobicity Analysis

Hydrophobicity affects protein structural stability, folding, and interactions with cellular components [?]. The average hydrophobicity H of each sequence was computed using the Kyte-Doolittle scale [?]:

$$H = \frac{1}{L} \sum_{i=1}^L H_i \quad (3.2)$$

where:

- H represents the average hydrophobicity of the sequence,
- L is the sequence length,
- H_i is the Kyte-Doolittle hydrophobicity score of the i^{th} amino acid.

These characteristics can help identify statistical outliers and reveal potential biases in the dataset.

3.3 Redundancy Reduction

Machine learning models trained on redundant sequences may become biased toward overrepresented patterns, reducing their generalizability. To mitigate this, redundancy reduction was applied to the dataset using MMseqs2 [16].

The redundancy reduction process involved the following steps:

1. The protein sequences, originally stored in FASTA format, were converted into an MMseqs2 database format to facilitate clustering.

2. An initial clustering step was performed using a sequence identity threshold of 100% . This step ensured that exact duplicate sequences were removed, preventing redundant entries in the dataset. The following command was executed:

```
mmseqs easy-cluster file.fasta dataset tmp -min-seq-id 1.0 -cov-mode 0 -c 1.0
```

3. After removing exact duplicates, a second clustering step was applied using a sequence identity threshold of 30%. This step grouped similar sequences into clusters. The following command was used:

```
mmseqs easy-cluster file.fasta dataset tmp -min-seq-id 0.3 -cov-mode 0 -c 1.0
```

This redundancy reduction step is implemented in the `redundancy-reduction.py` script.

3.4 Prediction of 3D Protein Folds

After applying redundancy reduction to the protein sequences, their three-dimensional (3D) structures were predicted using AlphaFold2 [reference?](#). The structural confidence of these predicted protein folds is quantified using the per-residue *predicted Local Distance Difference Test* (*pLDDT*) score. The pLDDT scores were used for exploratory purposes and no threshold was set based on pLDDT scores.

3.4.1 Extraction of Confidence Scores from AlphaFold2 Predictions

To evaluate the quality of the generated protein folds, a computational pipeline was developed to extract per-residue *pLDDT* scores from the predicted structures. This analysis was implemented using Python and the Biopython library that allows the parsing of Protein Data Bank (PDB) files. The processing pipeline consists of the following steps:

1. The program iterates through all PDB files generated by AlphaFold2, stored in the directory where each file corresponds to a single predicted protein structure.

2. The script employs the PDBParser module from Biopython to parse the structural information of each protein model. The pLDDT score is stored within the B-factor field of each atom in the PDB file. Since all atoms within a residue share the same pLDDT value, only one atom per residue is used to extract this confidence score.

3.5 Preparation of Training and Test Datasets

Two different methods were applied to create the training and test datasets, in order to utilize redundancy reduction to minimize sequence similarity biases and mitigating effects of data imbalance.

3.5.1 Method 1: Similarity-Based Data Splitting

After applying redundancy reduction with a sequence identity threshold of 30%, sequences that exhibited more than 30% similarity to any other sequence were excluded from the training set. This approach ensures that the training dataset consists of unique sequences. Sequences with more than 30% sequence similarity were designated as the test set. Conversely, sequences with less than 30% similarity were assigned to the training set, forming a non-redundant subset for model learning.

3.5.2 Method 2: Random Splitting After Redundancy Reduction

In the second approach, sequences that had more than 30% similarity were first removed from the training set, similar to Method 1. However, instead of directly assigning similar sequences to the test set, the remaining non-redundant sequences were randomly split into training and test sets.

A stratified random split was performed using the `train_test_split` function from Scikit-Learn 1.6.1, with a test size of 20% and a random state of 42 to ensure reproducibility:

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42)
```

3.6 Embedding Generation

~~In this study,~~ Protein embeddings were generated from both protein sequences and protein folds, utilizing protein language models: ProtT5 and ProST5.

For protein sequence based protein embedding generation, I employed the ProtT5 language model to generate embeddings for each protein sequence which encodes protein sequence features [17]. A Google Colab script called `embed_ProtT5.ipynb` was used during embedding generation. This script was used in November 2024.

To incorporate structural information, embeddings were also generated using the ProST5 language model from protein folds, which is trained to encode 3D structural features of proteins [18]. A Google Colab script called `ProST5.ipynb` was used during embedding generation. This script was used in December 2024.

3.7 Dimensionality Reduction

To reduce the complexity of embedding vectors while preserving essential information, Principal Component Analysis (PCA) was employed. PCA transforms the high-dimensional embeddings into a lower-dimensional space by keeping maximum variance [19].

This dimensionality reduction step is implemented in the `dimensionality-reduction.py` script.

3.8 Machine Learning Model Training and Evaluation

To ensure robust model validation, 5-fold cross-validation was applied, and Matthews Correlation Coefficient (MCC) was used as a primary metric for model evaluation.

3.8.1 Model Selection

Several machine learning algorithms were selected for the classification tasks. The models used in this study include: Citations

Dear Didem, the citations below are not working. Please check their existence in the bib file

- Random Forest [?]
- Logistic Regression [?]
- Support Vector Machine (SVM) [?]
- K-Nearest Neighbors (KNN) [?]

All machine learning models are saved in the `predictors` directory as `pickle files`

Can you add the extension associated with this type of files here, please?

.

3.8.2 Additional Approach: Hierarchical Predictor

To address class imbalances and improve classification accuracy, a hierarchical predictor was built. This approach first uses a predictor to predict between group exotoxins into broader type families based on their prevalence. A second-stage classifier is then applied within each group to classify specific exotoxin types with similar datapoints. The architecture of this predictor is shown in Figure 3.2 below.

3.8.3 Additional Approach: BLAST-Based Predictor

As a baseline method, BLAST (Basic Local Alignment Search Tool) was employed for sequence similarity analysis [?].

3.8.4 Negative Control: Scrambled Protein Sequence Embeddings

To verify that the machine learning models capture meaningful biological patterns, a negative control was implemented by generating embeddings from scrambled protein sequences. The shuffling of protein sequences was applied while preserving amino acid composition.

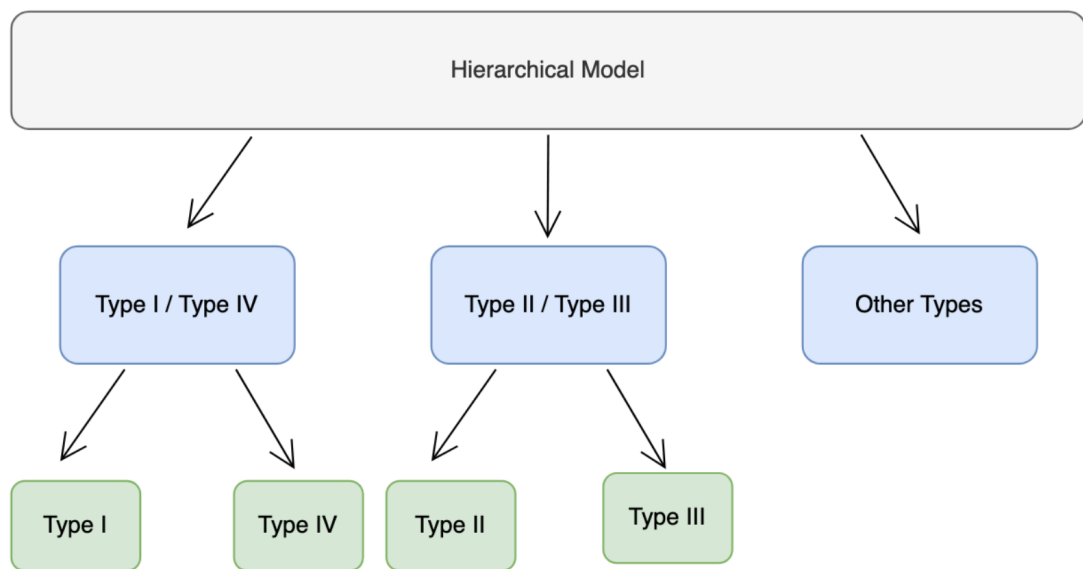


Figure 3.2: The hierarchical predictor architecture for exotoxin classification. The model employs a two-step approach to address class imbalance and improve classification accuracy. In the first stage, a high-level classifier groups exotoxins into broader type families based on their prevalence in the dataset. In ~~the~~ a second stage, a specialized classifier is applied within each group to refine the classification and distinguish specific exotoxin types that share similar features.

For this study, protein sequences were randomly shuffled while preserving their amino acid composition. These scrambled sequences were then processed through the same embedding pipeline using ProtT5 and ProstT5 to generate protein embeddings.

~~P~~ For detailed information, please refer to the link for the project's code and the structure of the project's data management plan: https://github.com/didemd/mlclassification_exotoxins.

Chapter 4

Results

This section presents two predictors, **ExoType** and **ExoTarget**, which differentiate bacterial exotoxin types and targets based on different inputs, including protein sequences ~~and~~ ~~or~~ ~~protein~~ ~~their~~ folds. ~~We~~ ~~I~~ first describe the data exploration and preprocessing steps, then outline the feature analysis, and finally discuss the model training and evaluation results.

4.1 Data Exploration and Preprocessing

In this section, we describe the data distribution among exotoxin types and targets and explain the preprocessing steps—including redundancy reduction—that prepare the dataset for modelling.

4.1.1 Dataset Overview

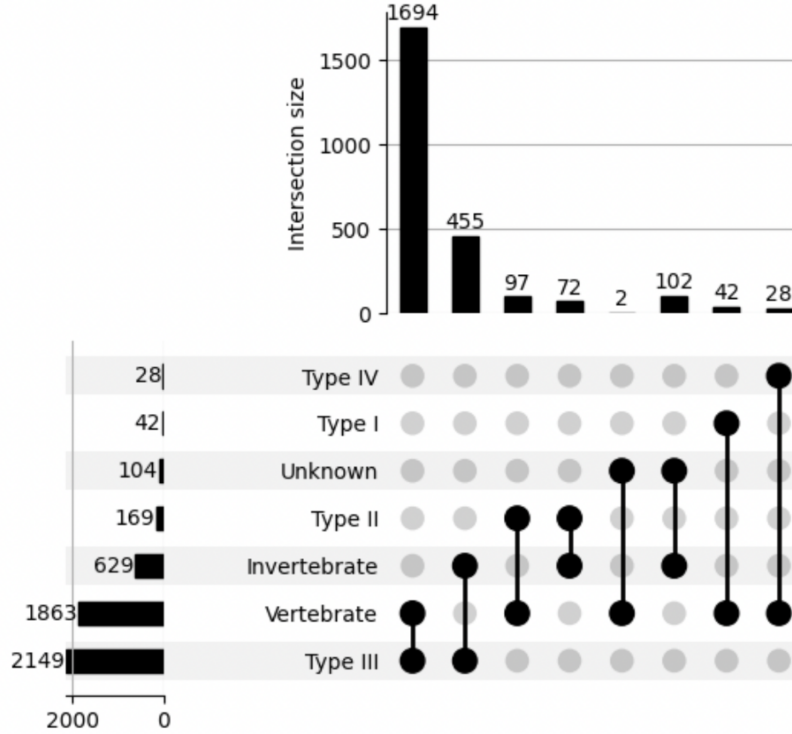


Figure 4.1: Upset plot illustrating the distribution of exotoxin types and their host targets. This plot visualizes the intersections between categories and highlights class imbalances in the dataset. The upper bar chart shows the intersection sizes, with the tallest bar (approximately 1,694) indicating the most common category intersection. The lower matrix, composed of black-filled circles on a grey background, maps each exotoxin type (Type I, Type II, Type III, Type IV, Unknown) and host target (Vertebrate, Invertebrate) to its intersection. The horizontal bars on the left are the total occurrences of each category.

Figure 4.1 summarizes the distribution of exotoxin types and targets within the dataset. Notably, the dataset is heavily skewed toward Type III exotoxins (2,149 sequences), which indicates a significant class imbalance that could affect model performance. In contrast, exotoxins categorized as "Unknown" comprise only 104 samples. Additionally, target classification shows that bacterial exotoxins targeting vertebrates (1,863 sequences) are three times more abundant than those targeting invertebrates (629 sequences). These

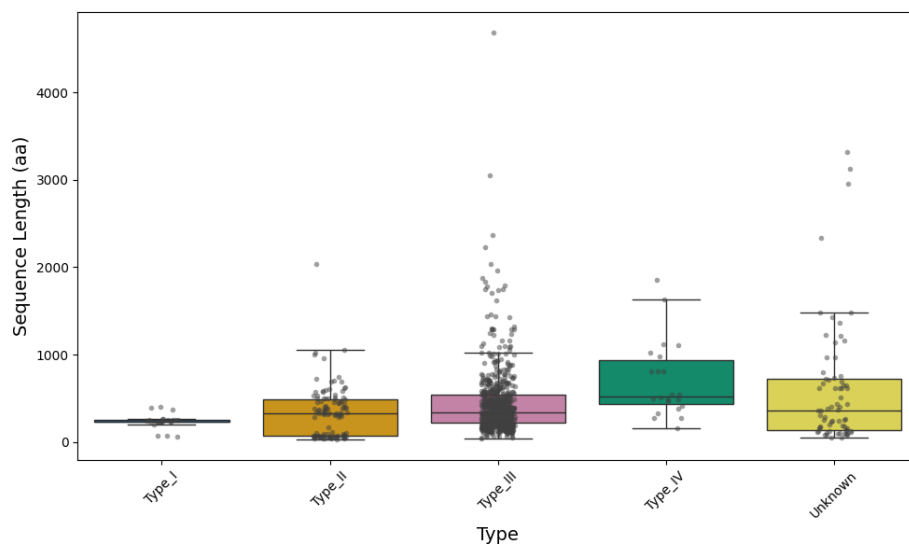
observations highlight the need to mitigate data imbalance, which will be observed in the following model training step.

4.1.2 Feature Analysis of Exotoxins

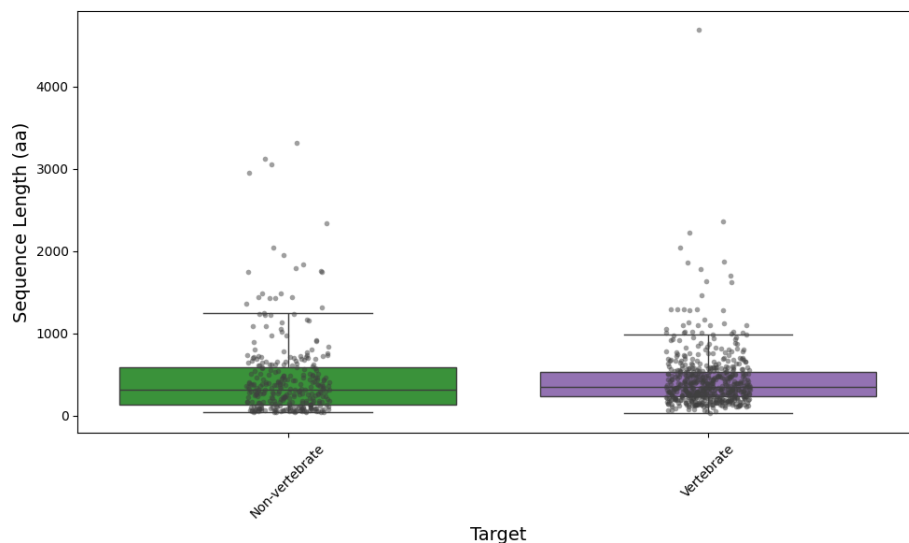
In this section, we inspect key sequence properties, such as sequence length and hydrophobicity, to examine potential biases within the dataset.

Sequence Length Analysis

A descriptive analysis of sequence lengths was conducted using boxplots generated from the dataset. Figure 4.2a displays the distribution of sequence lengths for each exotoxin type, while Figure 4.2b shows the distribution according to exotoxin targets.



(a)



(b)

Figure 4.2: Descriptive analysis of bacterial exotoxin feature: sequence length. (a) Sequence length distribution for different bacterial exotoxin types, measured in amino acids. The x-axis exhibit the exotoxin types, with each Type described by a colour: Type I (blue), Type II (orange), Type III (pink), Type IV (green), and Unknown (yellow). The y-axis shows sequence length in amino acids, ranging from 0 to approximately 3000, excluding three outliers. (b) Sequence length distribution for bacterial exotoxin targets, measured in amino acids. The x-axis represents the exotoxin targets, with each category represented by a distinct colour: Vertebrates (dark green) and Non-vertebrates (purple). The y-axis shows sequence length in amino acids, ranging from 0 to approximately 2500, excluding five outliers.

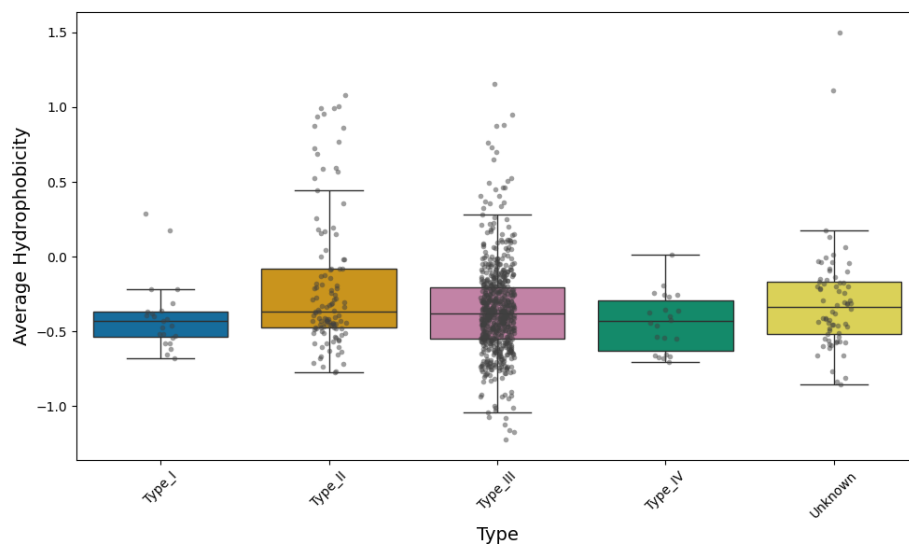
As observed in Figure 4.2a(a), the average sequence lengths differ across toxin types. Specifically, Type I toxins have a mean length of 239.87 amino acids, Type II toxins average 329.47 amino acids, Type III toxins measure approximately 448.01 amino acids, Type IV toxins have an average length of 706.00 amino acids, and toxins labelled as Unknown average 607.86 amino acids. In contrast, Figure 4.2b(b) reveals that non-vertebrate targets have a mean sequence length of 460.66 amino acids, whereas vertebrate targets average 438.74 amino acids.

A generalized linear model (GLM) was used to assess whether the differences in sequence length were statistically significant among the toxin types and targets. The GLM results indicate that toxin type is the primary factor affecting sequence length variability. In particular, compared to the overall average, Type I toxins are significantly shorter by 228.56 amino acids and Type II toxins are shorter by 136.04 amino acids, while Type IV toxins are significantly longer by 237.57 amino acids since all p values are lower than 0.002. The model did not show a statistically significant difference for Type III toxins, nor was there a significant effect of the target variable on sequence length.

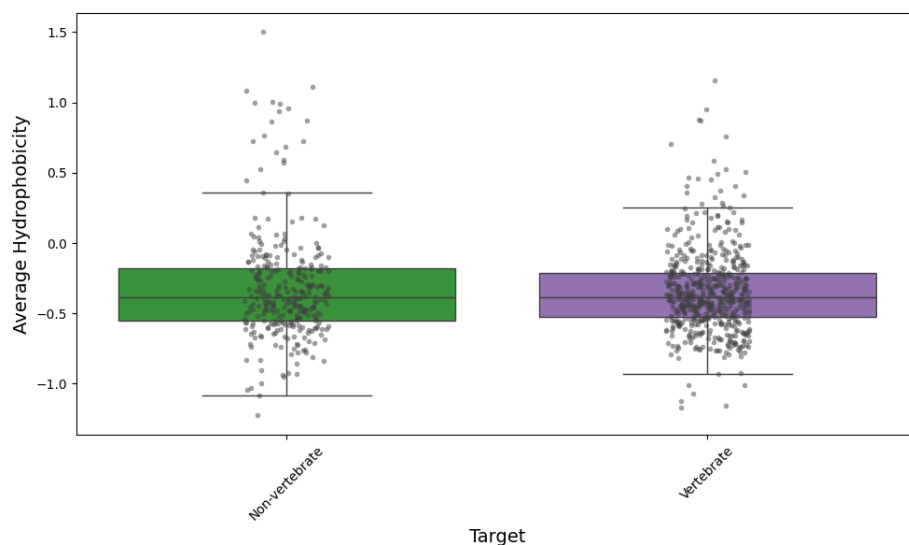
These findings suggest that the classification of exotoxins based on sequence length is largely driven by the toxin type rather than the biological target.

Hydrophobicity Feature Analysis

The hydrophobicity of bacterial exotoxins was analyzed per residue across different toxin types and targets to determine whether this feature systematically varies among groups. Figure 4.3a and Figure 4.3b visualize these distributions using box plots.



(a)



(b)

Figure 4.3: **Hydrophobicity distribution of bacterial exotoxin types and targets.** (a) The x-axis represents exotoxin types, with each type colour-coded: Type I (blue), Type II (orange), Type III (pink), Type IV (green), and Unknown (yellow). Box plots show the median, interquartile range, whiskers span to 1.5 times the interquartile range, with outliers as separate points. **Type II toxins exhibit the widest spread, while Type I and Type IV have more compact distributions.** (b) The x-axis represents exotoxin targets: Vertebrates (dark green) and Non-vertebrates (purple). **Hydrophobicity distributions are similar, though vertebrate-targeting exotoxins show slightly higher variability.**

In Figure 4.3a, Type I and Type IV toxins show more compact distributions, indicating lower variability, while Type II toxins exhibit the widest spread, suggesting greater diversity in sequence properties. The broader variation in Type II toxins may imply functional differences within this group. Type III and Unknown categories display moderate variability.

Figure 4.3b shows that vertebrate- and invertebrate-targeting exotoxins exhibit largely overlapping hydrophobicity distributions, with vertebrate-targeting toxins displaying slightly higher variability. However, the overlap suggests that hydrophobicity alone does not strongly differentiate exotoxin targets.

A Generalized Linear Model (GLM) analysis confirms that only Type II toxins show a statistically significant increase in hydrophobicity (coefficient: 0.1461, $p < 0.001$). No significant differences were found between target groups or other toxin types. These findings suggest that hydrophobicity may help distinguish certain toxin types, particularly Type II, but it is not a strong predictor for toxin targets.

4.2 Redundancy Reduction

This step was applied to ensure models ability to generalize unseen data.

First, MMseqs2 tool was used to set the similarity threshold to 100% in order to get rid of duplicates, if there are any. After that , 30% similarity was applied reduced the count from 2,492 to 1,068. This reduction reduces sequence similarity among dataset, allowing us to focus on the unique characteristics and functional diversity of exotoxins.

As shown in Table 4.1, after eliminating redundant sequences, the overrepresentation of Type III exotoxins remains even less pronounced (the percentage of Type III drops from 86% to 74%).

The redundancy reduction further limited the already scarce data, which increases the likelihood of the model memorizing it and overfitting. This, in turn, increases the necessity for dimensionality reduction later in the project pipeline. The drop in data imbalance will likely reduce bias toward the majority class in the final model. While reducing redundancy improves model generalizability, it further limits the available data, increasing the risk of overfitting.

Table 4.1: Summary of the dataset distribution based on exotoxin types (a) and their biological targets (b) before and after redundancy reduction.

(a)			(b)		
Exotoxin Type	Original	Reduced	Exotoxin Target	Original	Reduced
Type I	42	29	Vertebrates	1863	614
Type II	169	145	Invertebrates	629	234
Type III	2149	852			
Type IV	28	25			
Other Types	104	93			

4.2.1 Quality of Protein Folds Based On pLDDT Scores

Figure 4.4 displays the distribution of pLDDT scores for predicted nonredundant protein folds. This figure is used to evaluate the confidence of our protein structure predictions; higher scores indicate greater model confidence, while lower scores suggest regions that may require further analysis.

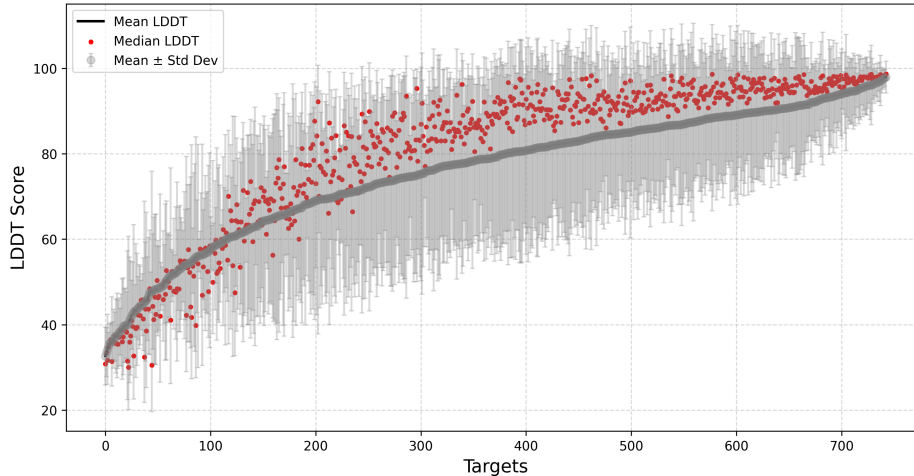


Figure 4.4: The black line is mean LDDT score across samples. The gray-colored region stands for one standard deviation around the mean, capturing the spread of LDDT scores within each sample. The red data points indicate the median LDDT scores, providing an alternative measure of central tendency and highlighting any deviations from the mean.

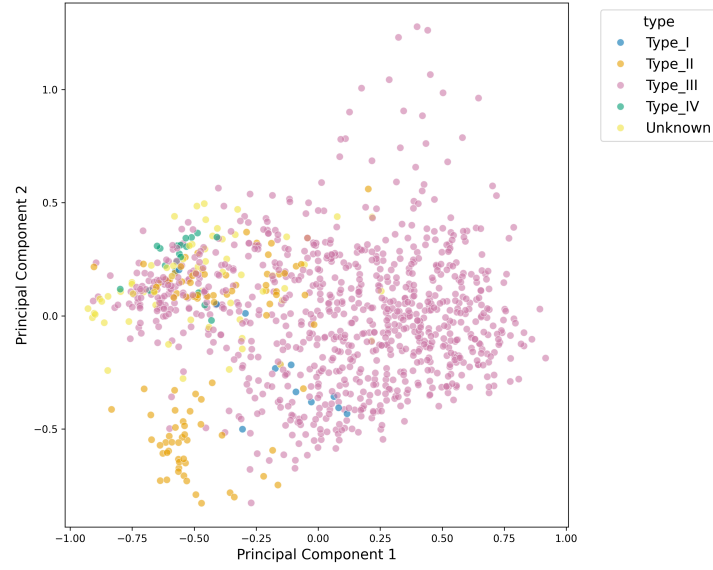
Approximately 68% of the data falls within the grey shaded area, corresponding to one standard deviation around the mean. The distribution shows that most LDDT scores lie between 40 and 100, with increasing variability as the mean score rises. Lower LDDT values (below 50) are present in a subset of samples, but the majority of structures show mean LDDT values above 60, suggesting overall reasonable confidence in the predictions. This analysis is useful in assessing the quality of predicted 3D structures before the prediction.

4.3 Dimensionality Reduction Techniques

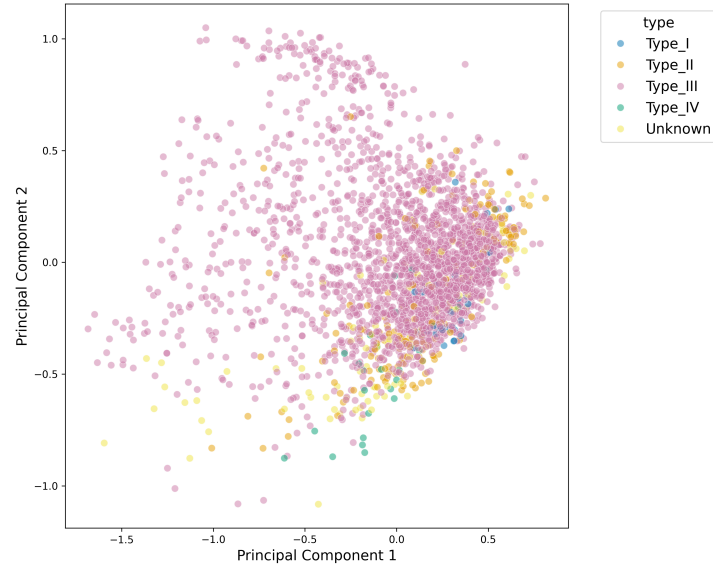
~~We~~ I applied Principal Component Analysis (PCA) to check the clustering patterns of bacterial exotoxins and their targets using both regular and scrambled protein embeddings. The results reveal important insights into the role of sequence-based embeddings in toxin classification.

The PCA of non-scrambled embeddings (Figure 4.5a) shows clustering among toxin types.

The PCA of scrambled embeddings (Figure 4.5b) reveals no meaningful structure, with all toxin types overlapping substantially. This lack of clustering indicates that the scrambled embeddings fail to capture meaningful toxin-specific information, as opposed to the regular embeddings.



(a) (a)



(b) (b)

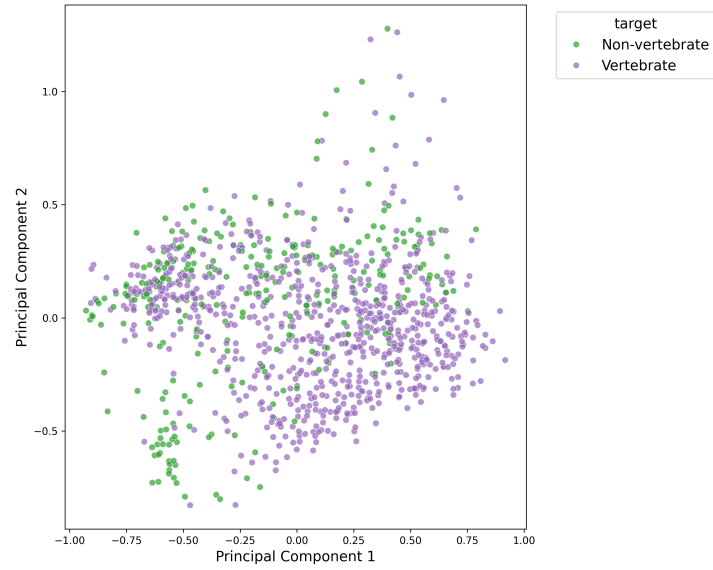
Figure 4.5: (a) Principal Component Analysis (PCA) projection of bacterial exotoxin types Each exotoxin type is colour-coded: Type I (blue), Type II (purple), Type III (pink), Type IV (green), and Unknown (yellow) (b)The scrambling reduces separation among the clusters, validating the effect of the shuffling of protein sequences.

Figure

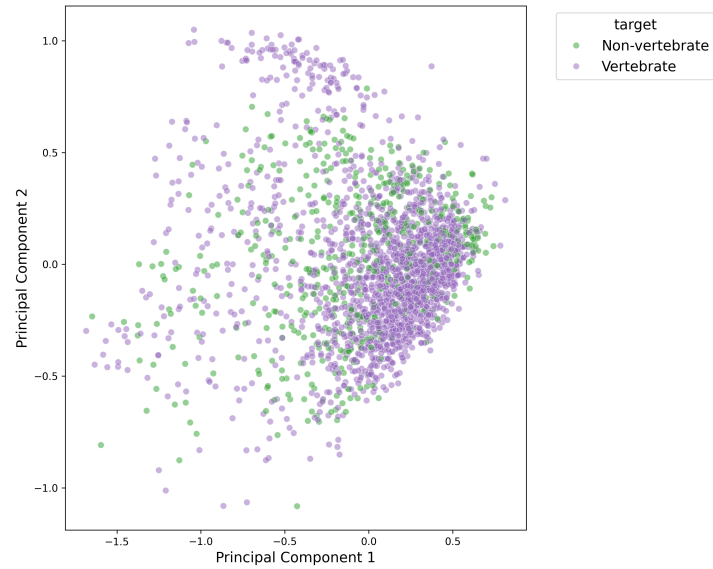
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Similarly, Figure 4.6a and Figure 4.6b show the PCA projections for exotoxin targets, confirming that vertebrate and invertebrate sequences exhibit biologically meaningful clustering.

The PCA of regular embeddings (Figure 4.6a) reveals small clustering patterns among vertebrate toxin targets. The PCA of scrambled embeddings (Figure 4.6b) reveals no meaningful structure, with all toxin targets overlapping.



(a)



(b)

Figure 4.6: Principal Component Analysis (PCA) projection of bacterial exotoxin targets and scrambled PCA projection of bacterial exotoxin targets. The x-axis describes Principal Component 1, and the y-axis describes Principal Components Targets are color-coded: Vertebrates (purple) and Non-vertebrates (dark green). Plus, the scrambling reduces cluster compactness, demonstrating the loss of structure in the data.

4.4 ExoType Predictor

This section describes the development of the ExoType predictor, which classifies exotoxins into five categories: Type I, Type II, Type III, Type IV, and Unknown. The following subsections outline the training and evaluation of the model using two different data representations:

1. Sequence-based embeddings (generated from protein sequences using MMseqs2)
2. Fold-based embeddings (derived from protein structures predicted by AlphaFold2)

4.4.1 Model Training

Five models with different architectures were trained to predict different exotoxin types: Logistic Regression, Support Vector Machines, K-Nearest Neighbours, Hierarchical Model, and Random Forest.

- Logistic Regression,
- Support Vector Machines,
- K-Nearest Neighbours,
- Random Forest,
- Hierarchical Model

To evaluate performance, we applied two data-splitting strategies:

Approach 1: Standard Train-Test Split

The dataset of 1,067 embeddings was divided into:

- 80% training set (848 samples)
- 20% test set (213 samples)

Class distribution in the training set:

- **Type I:** 10
- **Type II:** 77

- **Type III:** 699
- **Type IV:** 14
- **Unknown:** 48

Approach 2: Redundancy Reduction Based Split

The dataset of 1,067 embeddings was divided into:

- **Type I:** 12
- **Type II:** 96
- **Type III:** 875
- **Type IV:** 18
- **Unknown:** 60

The best cross-validation MCC scores for each model under both data-splitting approaches are summarized in Table 4.2. **MCC scores**

When using a table with numbers, you still have to include the confidence interval or the SEM for the values. Usually is give as 0.62(+/- 0.03) and the Column should have the unit (Standard split (SEM) or Standard split (95%CI))

Table 4.2: MCC Scores for Different Models

Model	Standard Split	Redundancy Reduction Split
Random Forest	0.62	0.62
Logistic Regression	0.62	0.64
SVM	0.63	0.65
KNN	0.74	0.73

Comparing the two approaches reveals several key differences. In the original split, the training set consisted of 848 samples with relatively fewer samples for the minority classes, whereas the redundancy reduction approach yielded a training set with slightly higher counts for these classes, yet the overall distribution remained heavily skewed toward Type III.

Next paragraph

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In Figure ?? In terms of cross-validation MCC, the original approach achieved values of 0.62, 0.62, 0.63, and 0.74 for Random Forest, Logistic Regression, SVM, and KNN respectively. With the second approach, Random Forest had an MCC of 0.62, while Logistic Regression and SVM improved to 0.64 and 0.65, respectively, and KNN decreased to 0.73. These changes mean that while the redundancy reduction approach has improved the performance of some models, due to the increasing of the minority classes, the overall impact is similar. The class imbalance in the second approach may limit the models' ability to generalize better, indicating that while there is a hint of improvement, additional strategies to further address the imbalance might be necessary to prevent the potential underfitting of the minority classes.

In addition to the numerical comparisons, the learning curves provide further insight into how each model responds to increasing amounts of training data under both approaches. In Figure 4.7, the learning curves for these four models display the training and validation MCC scores across various training set sizes. Ideally, as the amount of training data increases, the training MCC should decline slightly—indicating that the model stops overfitting and starts to generalize better to unseen data.

For Random Forest and KNN, the training MCC remains nearly constant, suggesting these models continue to overfit the training data. Conversely, SVM and Logistic Regression exhibit lower training MCC scores, implying they do not overfit. Moreover, although the validation MCC scores for all models improve with more data, SVM and Logistic Regression appear more balanced between fitting the training data and generalizing well. This pattern indicates that SVM and Logistic Regression are more suitable for the classification of toxin types using sequence-based embeddings.

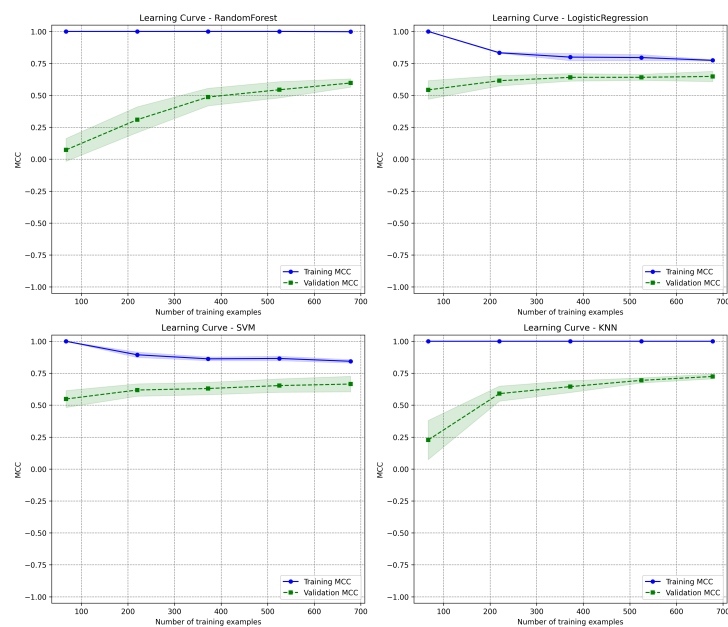


Figure 4.7: MCC Bar Plot for Protein Sequence-Based Embeddings: This graph is used to assess model generalization when the dataset is split into 80% training and 20% test sets.

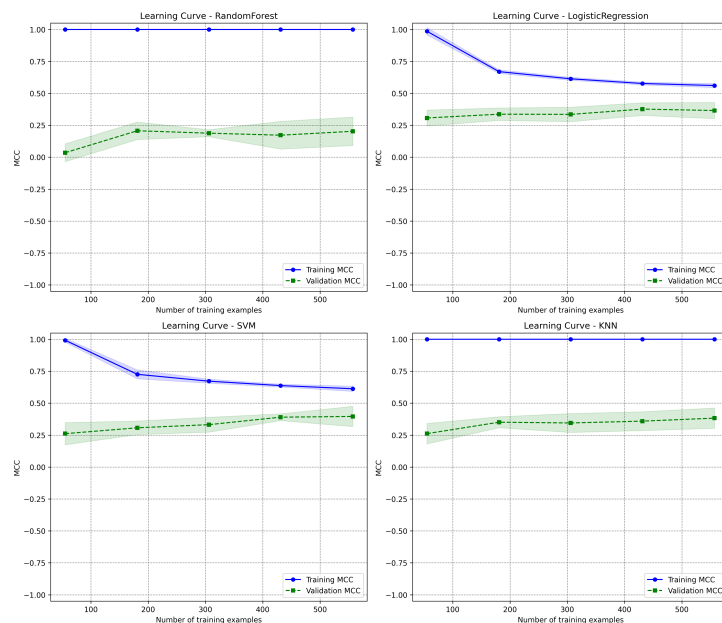


Figure 4.8: MCC Bar Plot for Protein Fold-Based Embeddings: This graph is used to assess model generalization when the dataset is split into 80% training and 20% test sets.

In Figure 4.8, the learning curves for the protein folds illustrate how each classifier’s MCC on the training and validation sets evolves as more training samples become available. Random Forest shows a relatively stable performance but remains at a low validation MCC of 0.16 in cross-validation, suggesting that while it fits the training data quickly, it struggles to generalize. Logistic Regression (0.36), SVM (0.37), and KNN (0.38) exhibit modest improvements in validation MCC with increasing training size, but none converge on particularly high MCC values. These trends underscore the challenge posed by class imbalance and potential redundancy in the data, as well as the need for more extensive feature engineering or alternative approaches to further boost model generalization.

4.4.2 Model Performance Evaluation

This section evaluates the performance of five machine learning models trained on two different embedding types: sequence-based embeddings and protein fold-based embeddings. The models include Logistic Regression (LR), Sup-

port Vector Machines (SVM), K-Nearest Neighbors (KNN), Random Forest (RF), and Hierarchical Models (Hier_RF, Hier_SVM). Additionally, BLAST, a non-machine learning baseline based on sequence similarity, is included for comparison.

Performance assessment is based on the Matthews Correlation Coefficient (MCC), with error bars indicating standard deviation across cross-validation folds. Learning curves are also analyzed to detect potential overfitting.

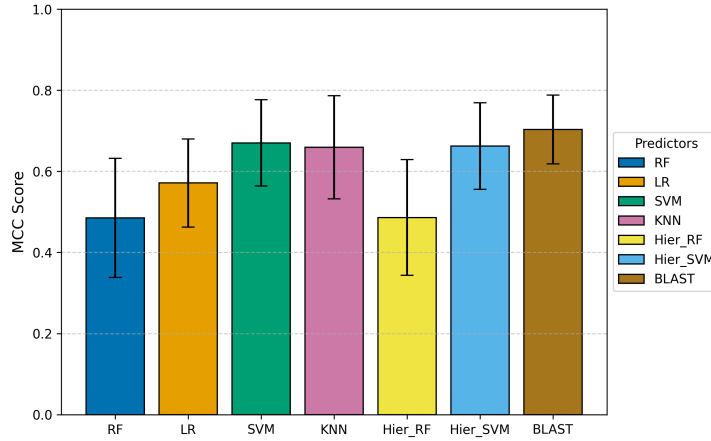


Figure 4.9: MCC Bar Plots for Protein Sequence-Based Embeddings. The x-axis represents different classification models: Random Forest (RF, blue), Logistic Regression (LR, orange), Support Vector Machine (SVM, green), K-Nearest Neighbors (KNN, pink), Hierarchical Random Forest (Hier_RF, yellow), Hierarchical Support Vector Machine (Hier_SVM, light blue), and BLAST predictor (brown). The y-axis indicates the Matthews Correlation Coefficient (MCC) scores, with higher values reflecting better classification performance. Error bars represent the standard deviation across cross-validation folds. (a) The first MCC Plot corresponds to the first approach, where the dataset was split into 80% training and 20% test sets.

Figure 4.9 presents the MCC scores for models trained on **sequence-based embeddings**. SVM and KNN achieved the highest MCC scores, approximately 0.67 and 0.68 in the first approach (80% training, 20% test). Their performance improved further with a larger training set, exceeding 0.80 in the second approach. Random Forest and BLAST showed moderate performance MCC of 0.60 in the first approach, increasing to 0.75 in the second one. On the other hand, Logistic Regression and Hierarchical Random

Forest had the lowest MCC scores, remaining below 0.50.

To see if the model experiences overfitting or not, **learning curves** were inspected. As a result, it can be seen that Random Forest and KNN have high training MCC values, suggesting overfitting. SVM and Logistic Regression had lower training MCC scores, indicating better generalization.

Overall, SVM and KNN were the most effective classifiers for sequence-based embeddings. Increasing the training data improved MCC scores across all models, highlighting the impact of dataset size on model generalization.

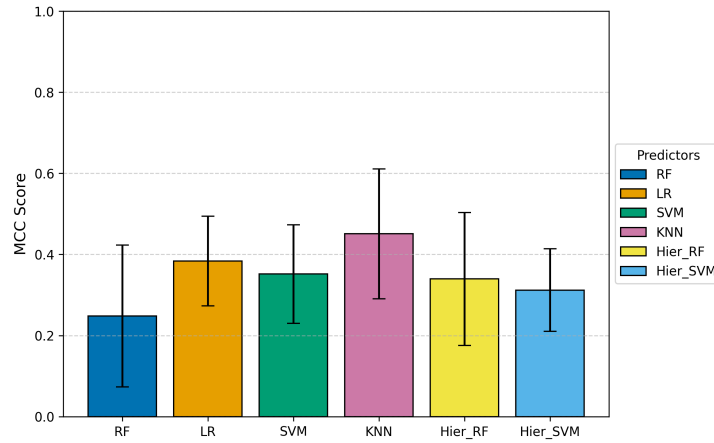


Figure 4.10: MCC Bar Plot for Protein Fold-Based Embeddings. The x-axis represents different classification models: Random Forest (RF, blue), Logistic Regression (LR, orange), Support Vector Machine (SVM, green), K-Nearest Neighbors (KNN, pink), Hierarchical Random Forest (Hier_RF, yellow), Hierarchical Support Vector Machine (Hier_SVM, light blue), and BLAST predictor (brown). The y-axis indicates the Matthews Correlation Coefficient (MCC) scores, with higher values suggesting better classification performance. ~~(a) The first MCC Plot belongs to the first approach while (b) the Second MCC Plot belongs to the second approach, where the whole nonredundant dataset was used for training, and previously removed redundant data was used as the test set. Models trained using the second approach generally show higher MCC scores, demonstrating the benefit of increased training data. The hierarchical models (Hier_RF, Hier_SVM) exhibit improved MCC scores, highlighting their ability to mitigate class imbalance.~~

Figure

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Figure 4.10 presents the MCC values for predictors built using protein fold-based embeddings. It was visible that fold-based embeddings performed worse than sequence-based embeddings due to its MCC values. Among the models, KNN achieved the highest MCC score, around 0.50, and Logistic Regression with an MCC value of 0.40. The hierarchical models provided some improvements, but their effectiveness varied across different test scenarios. The error bars in the figure indicate higher variance in MCC scores, suggesting that models trained on fold-based embeddings exhibited greater instability across cross-validation folds.

The lower performance of models on fold-based embeddings suggests that sequence-based representations may be more effective for exotoxin classification. These results highlight the importance of embedding choice in exotoxin classification. Sequence-based embeddings consistently led to higher MCC scores, particularly when used with SVM, KNN, and hierarchical models. However, fold-based embeddings still provide valuable structural information.

Evaluation of Classifier Performance: ROC and Precision-Recall Curves

To quantitatively assess classifier performance, we analyze two key evaluation metrics: the **Receiver Operating Characteristic (ROC) curves**

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and **Precision-Recall (PR) curves**. These metrics are particularly crucial for imbalanced datasets, as accuracy alone may provide misleading results.

Figures 4.11 and 4.13 display the Receiver Operating Characteristic (ROC) and Precision-Recall (PR) curves, respectively, for models trained on sequence-based embeddings, while Figures 4.12 and 4.14 present the corresponding curves for models trained on protein fold-based embeddings. These evaluation metrics are essential for assessing classification performance, particularly in imbalanced datasets, where accuracy alone can be misleading.

Receiver Operating Characteristic (ROC) **figures ROC**

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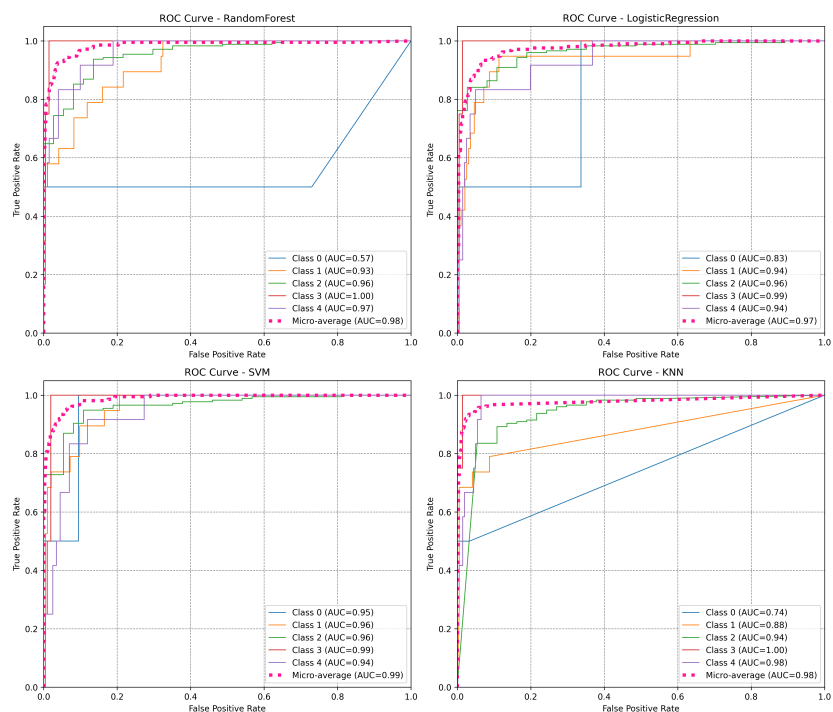


Figure 4.11: ROC Curve for Protein Sequence-Based Embeddings. This curve is used to evaluate the sensitivity and specificity for a split test.

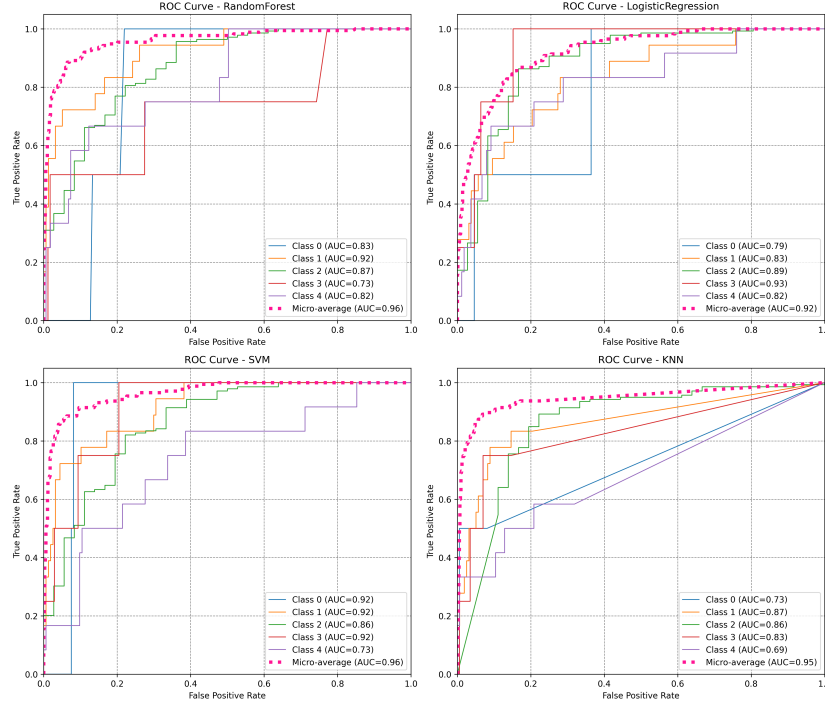


Figure 4.12: ROC Curve for Protein Fold-Based Embeddings.

Figure 4.11 presents the ROC curves for multiple classification models trained on sequence-based features, evaluating their performance under different testing conditions. The results indicate that SVM and Random Forest achieved the highest AUC scores (0.98 and 0.99, respectively), demonstrating strong generalization. Logistic Regression also performed well (AUC 0.97) but exhibited slightly lower sensitivity for certain classes. In contrast, KNN showed inconsistencies across classes, with an AUC of 1.00 for Type III but only 0.74 for Type I, indicating variability in its class separability.

Figure 4.12 illustrates the ROC curves for models trained on protein fold-based embeddings. The overall AUC values were lower than in sequence-based embeddings. Despite this, SVM and Random Forest still performed well (AUC 0.96 and 0.92, respectively), albeit at slightly reduced levels. Logistic Regression and KNN struggled more in this setting, showing reduced class separability. Plus, the increased variance in AUC scores suggests lower classification performance when using fold-based embeddings as raw data.

Precision-Recall Curves Figure 4.13 illustrates the PR curves for models trained on sequence-based embeddings. The results align with the ROC analysis, showing that Random Forest and SVM outperform other classifiers. Both models maintain high precision across different recall levels, leading to micro-average AP scores of 0.96 and 0.94, respectively. Logistic Regression also exhibits competitive performance, though with slightly more variability, as indicated by a micro-average AP of 0.93. KNN, on the other hand, performs inconsistently, with some classes displaying drops in precision, particularly at higher recall levels, further confirming its instability in this feature space.

Figure 4.14 displays the PR curves for models trained on fold-based embeddings. The micro-average AP scores for all models decrease compared to sequence-based embeddings, confirming that classification is more challenging in this setting. The PR curves reveal significant fluctuations in precision for Logistic Regression and KNN, with KNN experiencing the most dramatic drops. Random Forest and SVM remain the most robust models, although their AP scores are reduced compared to the sequence-based setting (Random Forest AP = 0.90, SVM AP = 0.91). These observations indicate that fold-based embeddings may lead to a higher false positive rate, reducing precision in classification tasks.

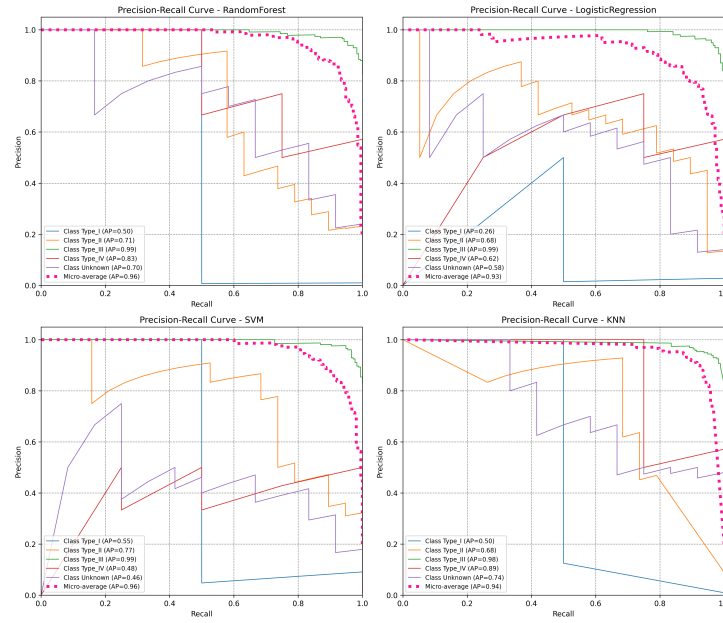


Figure 4.13: Precision-Recall Curves for Protein Sequence-Based Embeddings. These curves are used to assess the balance between precision and recall across different models.

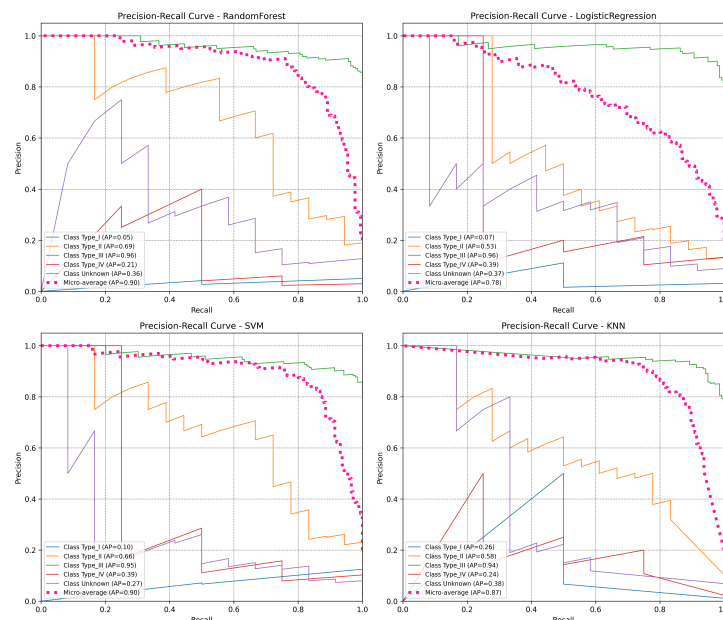


Figure 4.14: Precision-Recall Curve for Protein Fold-Based Embeddings. This plot is used to evaluate the precision-recall trade-off for models using protein fold-based features.

Overall, the comparison between Figures 4.13 and 4.14, sequence-based embeddings lead to better classification performance, while fold-based embeddings introduce challenges that make classification more difficult. A comparison between Figures 4.13 and 4.14

4.4.3 ExoTarget Predictor

This section describes the development of the ExoType predictor, which classifies exotoxins into two categories: Vertebrates and Non-vertebrates. The following subsections outline the training and evaluation of the model using two different data representations:

1. Sequence-based embeddings (generated from protein sequences using MMseqs2)
2. Fold-based embeddings (derived from protein structures predicted by AlphaFold2)

4.4.4 Model Training

Five models with different architectures were trained to predict different exotoxin types: Logistic Regression, Support Vector Machines, K-Nearest Neighbours, and Random Forest.

- Logistic Regression,
- Support Vector Machines,
- K-Nearest Neighbours,
- Random Forest,

To evaluate performance, we applied two data-splitting strategies: The dataset of 1,067 embeddings was divided into:

- 80% training set (848 samples)
- 20% test set (213 samples)

Class distribution in the training set:

- **Vertebrates:** 614
- **Invertebrates:** 234

Approach 2: Redundancy Reduction Based Split

Class distribution in the training set:

- **Vertebrates:** 768
- **Invertebrates:** 293

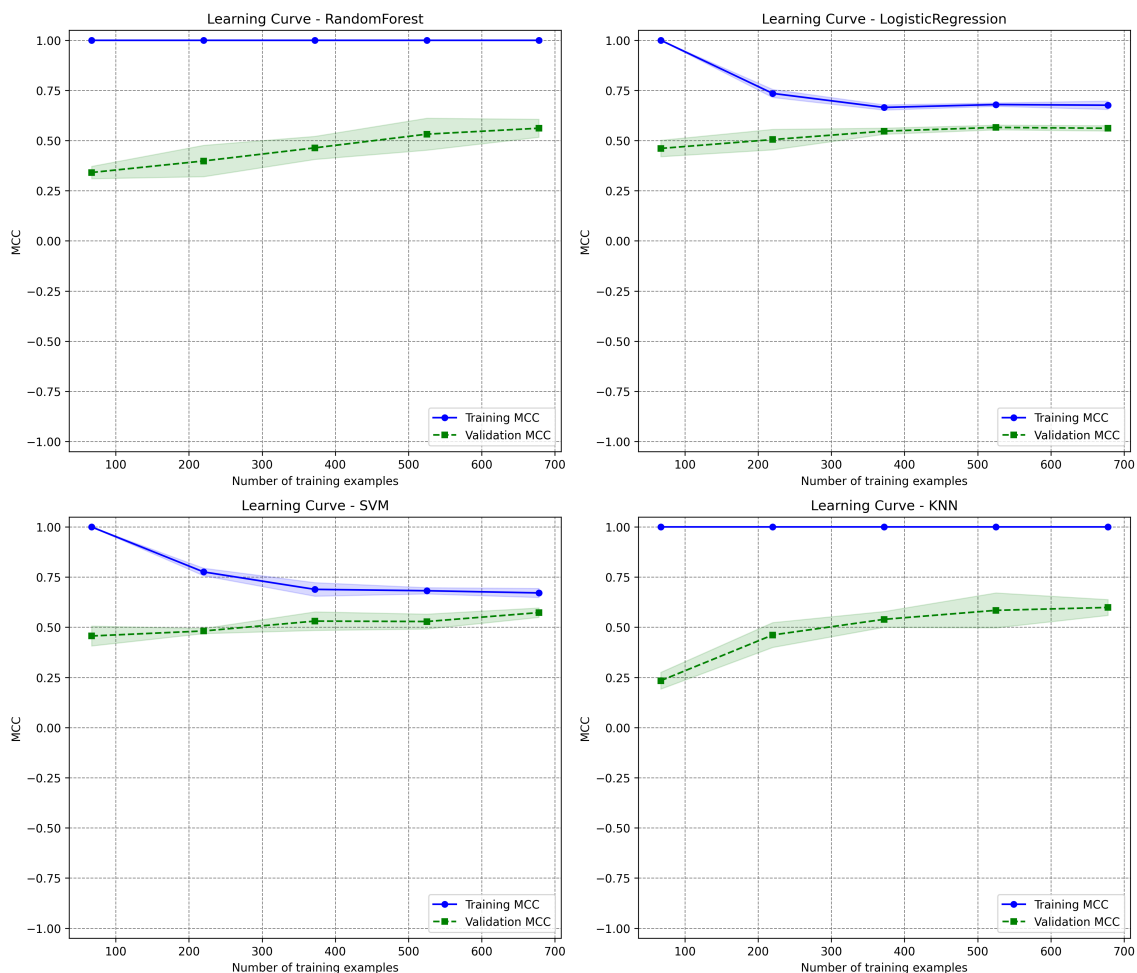


Figure 4.15: Learning Curves for Protein Sequence-Based Embeddings

In Figure 4.15 illustrate how the MCC changes for the training and validation sets as the training sample size increases. For Random Forest, the training MCC starts on 0.80 at 100 samples and stabilizes on 0.72 at 700 samples, while the validation MCC improves from 0.68 to 0.72. Logistic Regression's training MCC rises from about 0.72 to 0.68, with validation of 0.65. SVM begins around 0.80 for training and ends around 0.75, while validation moves between 0.65 to 0.72. KNN's training curve decreases from 0.78 to 0.73, resulting with validation around 0.68.

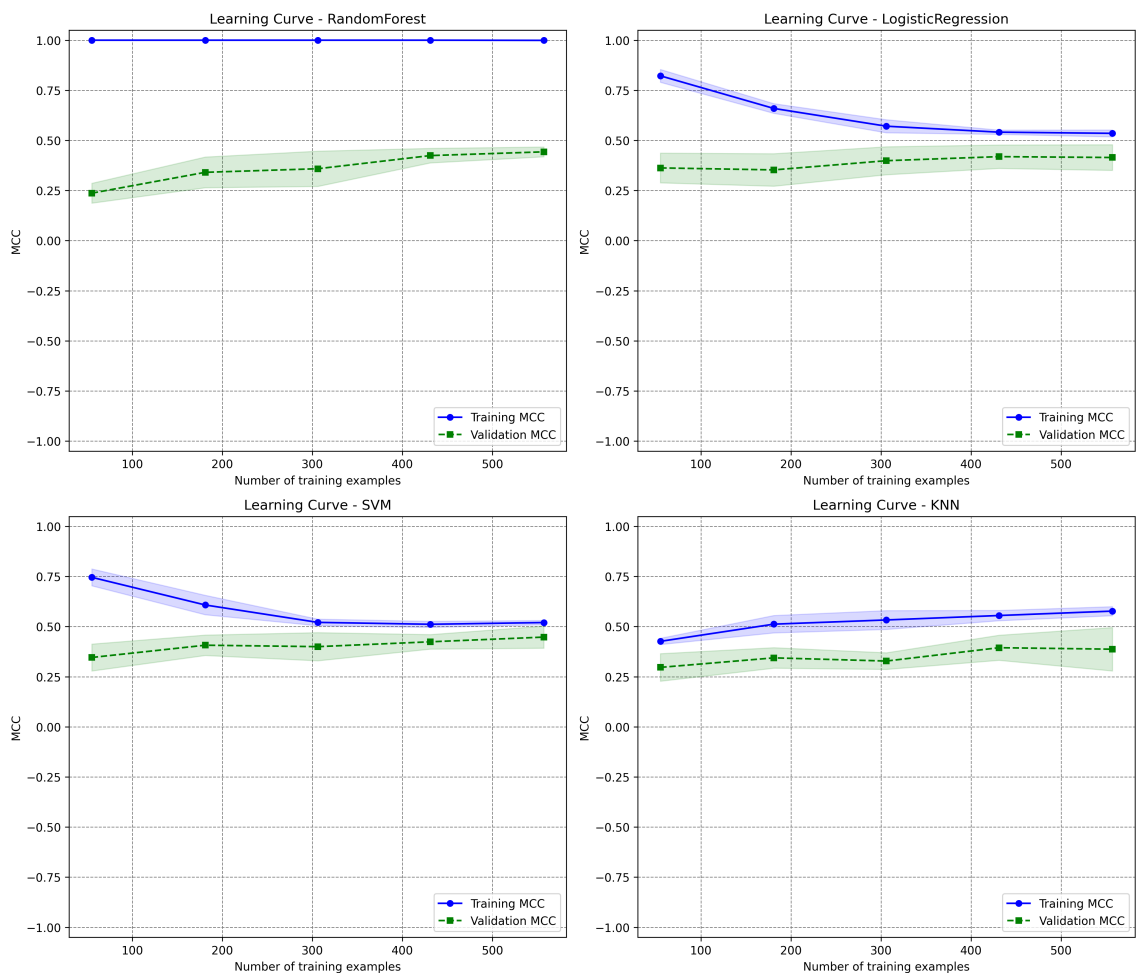


Figure 4.16: Learning Curve for Protein Fold-Based Embeddings

In Figure 4.8, the learning curves for the protein folds illustrate how each classifier's MCC on the training and validation sets evolves as more training samples become available. Random Forest's MCC is around 0.75 for the first approach and improves to around 0.78, with validation reaching 0.73. Logistic Regression starts around 0.70 and stabilizes, while its validation MCC increases to 0.69. SVM starts around 0.75 for training and moves to about 0.78, with validation up to 0.72. KNN's training MCC increases slightly from 0.68 to 0.73, while the validation MCC improves from 0.62 to 0.68.

4.4.5 Performance Evaluation

MCC Bar Plots

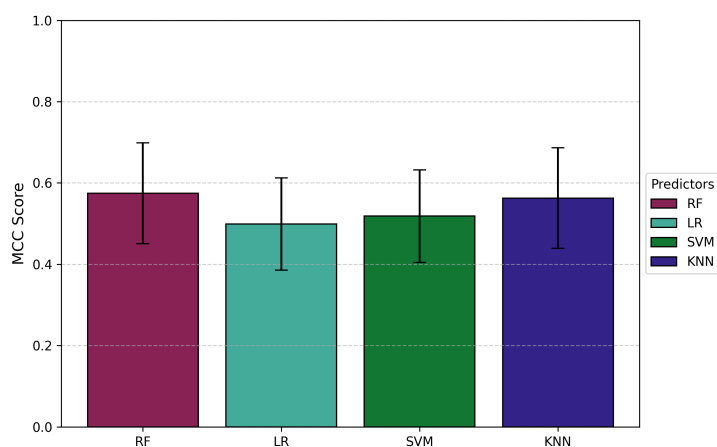


Figure 4.17: MCC Bar Plot for Protein Sequence-Based Embeddings

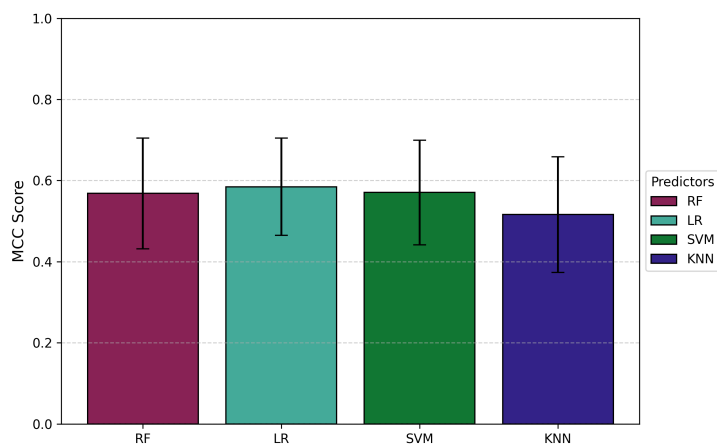


Figure 4.18: MCC Bar Plot for Protein Fold-Based Embeddings

In Figure 4.17 and In Figure 4.18 display the final MCC values for each model under sequence-based and fold-based embeddings. For the sequence-based embeddings, Random Forest achieves approximately 0.77, Logistic Regression around 0.73, SVM around 0.74, and KNN about 0.71. For the fold-based

embeddings, Random Forest MMC is 0.72, Logistic Regression at 0.70, SVM at 0.71, and KNN at 0.68.

ROC Curves

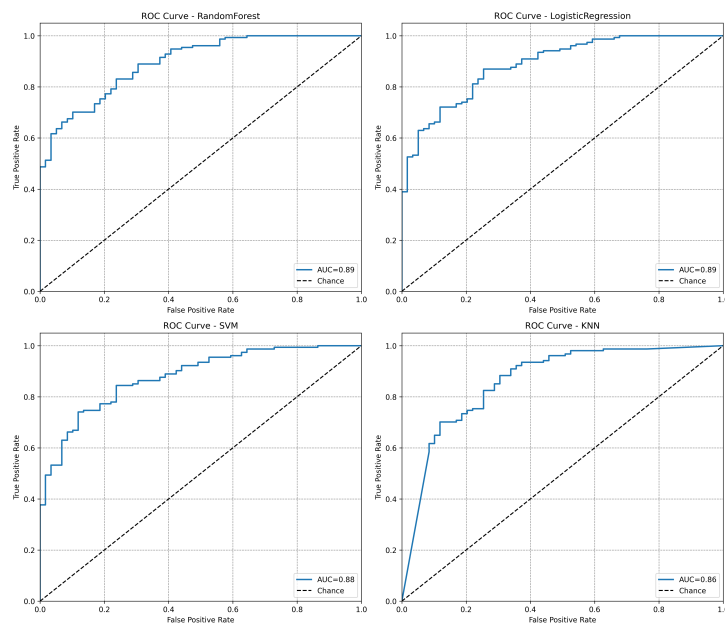


Figure 4.19: ROC curve Plot for Protein Sequence-Based Embeddings

The ROC curves provide an overview of how each classifier separates the two exotoxin classes at various thresholds. With sequence-based embeddings in Figure 4.19 Random Forest has an approximate AUC of 0.91, Logistic Regression about 0.88, SVM around 0.90, and KNN is 0.87.

Chapter 5

Conclusion

In this chapter, we conclude from the findings presented in this thesis on the machine learning classification of bacterial exotoxins. We summarize our key findings, discuss the limitations of our approach, and outline potential directions for future research based on the insights gained from our study.

5.1 Summary

Our research has made several contributions to the classification and characterization of bacterial exotoxins using machine learning. Key findings include:

- **ExoType and ExoTarget Predictors:** We developed machine learning-based tools, ExoType and ExoTarget, for classifying bacterial exotoxin types and predicting their specific targets. Our results show that these models can effectively distinguish between toxin types and targets with significant Matthews Correlation Coefficient (MCC) values.
- **Protein Embeddings for Improved Classification:** We explored different protein representations, including primary sequences and protein embeddings derived from ProtT5 and ProstT5 models. Our findings indicate that embeddings preserve crucial biochemical properties and enhance classification performance.
- **Dimensionality Reduction and Redundancy Elimination:** Principal Component Analysis (PCA) to reduce the dimensionality of protein embeddings from 1024 to 50 dimensions while retaining 90% of

the variance. Additionally, we performed redundancy reduction using MMseqs2 to minimize sequence similarity bias, reducing the dataset size while maintaining diversity.

- **Hierarchical Predictor as an Alternative Approach:** Besides conventional classifiers such as Random Forest and Support Vector Machines (SVM), we developed a hierarchical predictor approach that classifies exotoxins in a stepwise manner, reducing overfitting and improving performance for underrepresented classes.

5.2 Limitations

During this thesis project, there were many challenges that deserves recognition:

- **Small Dataset Size:** The dataset used in this thesis project was quite small, around 2400 datapoints, which can affect the generalizability of the models.
- **Data Imbalance:** The dataset had significant class imbalance, particularly for some exotoxin types such as overrepresentation of Type III and Vertebrate labels. This resulted in models favoring overrepresented classes, potentially affecting generalizability.
- **Lack of Extensive Benchmarks:** There is a limited number of external benchmarks available for evaluating exotoxin classification models, making comparative analysis challenging.
- **Black-Box Nature of Embeddings:** Although protein embeddings capture structural and biochemical properties, their interpretability remains a challenge, as they do not directly correspond to human-readable sequence features.

5.3 Future Work

Building on the foundation set by this thesis, several promising areas for future research can be outlined:

- **Integration of Biochemical Features:** Future work could incorporate biochemical properties such as aromaticity and isoelectric point into machine learning models.
- **Development of a User-Friendly Interface:** A interface for ExoType and ExoTarget would provide easier use by researchers and experimentalists, allowing them to input protein sequences and receive predictions without requiring extensive computational expertise.
- **Collaboration with Wet-Lab Research:** Validating our machine learning predictions through experimental studies would be essential by the aid of ground-truth data.

In conclusion, this thesis investigated methods to classify bacterial exotoxins using machine learning. Our contributions include the development of ExoType and ExoTarget predictors, the exploration of protein embeddings. While challenges such as data imbalance still exists, our findings provide a foundation for future advancements in the computational classification of bacterial exotoxins.

Chapter 6

Discussion

Appendix

6.1 Additional Performance Metrics

This section provides additional performance metrics, including Matthews Correlation Coefficient (MCC), Precision, Recall, and F1-score.

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